Assignment 1

ADVANCED SIMULATION

EPA133A

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1 Data Quality Diagnosis

In this chapter, the data quality issues identified in the datasets are analyzed and categorized. By applying the frameworks of Huang (2013) regarding data quality criteria and Marsden (2018) regarding the context of data generation, we can better understand the technical and procedural roots of these errors.

1.1 Road Dataset

In the roads dataset, several data quality problems were found. This dataset uses Location Reference Points (LRPs) to draw the roads in Bangladesh, where the road is defined as the minimum length line that connects all LRPs. We identified several type of data quality issues.

1.1.1 Spatial Outliers and Coordinate Errors

The most serious data quality problem is the presence of extreme spatial outliers. From a Syntactic perspective [2], these represent a failure in Syntactic Accuracy, where the data values do not conform to the expected domain for Bangladesh's geographic coordinates.

In a clean dataset, the distance between consecutive LRPs should be fairly consistent within a road, as conditions stay relatively the same along that road. There may be some large gaps due to variance, but inspection reveals that there is rarely a 'correct' LRP with a large distance from both its neighbours. However, the raw dataset contains points that are located far away from their neighbouring points, and disrupt the natural flow of the road.

When the roads are visualized, these errors create strange distortions. A road segment that should only be a few kilometers long, sometimes suddenly jumps 100 kilometers away. In some cases, the road even appears to move into the ocean or into another country before returning to the correct path at the next LRP. In terms of Semantics, this represents a lack of semantic accuracy, as the recorded value deviates from the real-world value.

1.1.2 Causes

These problems are usually caused by human data entry mistakes, following the '7 Why's framework' [3], these problems are primarily rooted in the who and how of data collection. Common causes include:

- Decimal errors: a decimal point is placed incorrectly (for example, writing 90.45 as 904.5). This moves the coordinate far away from its correct position.
- Swapped coordinates: latitude and longitude values are accidentally switched, placing the point in the wrong location.
- Typing mistakes: simple typing errors, such as switching digits or entering incorrect numbers during data collection.

1.1.3 Topological Inconsistencies

Besides extreme outliers, there are also problems related to the logical order of the road points. A road is a continuous line, which means that point LRP_n should be followed geographically by LRP_{n+1}.

In some cases, the data shows 'zigzagging' or 'backtracking'. This means that a point appears to move backwards instead of forward along the road. Although roads can curve naturally, sharp backward movements are unrealistic. This indicates a failure in Syntactic Consistency [2], suggesting the LRPs are not correctly sorted. Marsden (2018) would categorize this as a 'How' issue. Overall, this may indicate that the LRP's are not sorted correctly in the dataset. It could also be caused by small coordinate errors, such as GPS inaccuracies.

1.2 Bridges Dataset

1.2.1 Spatial Outliers and Coordinate Errors

Applying the data quality framework of Marsden and Pingry (2018), the bridges dataset reveals several weaknesses in the What, Where, and How dimensions of data collection. When visually inspecting the bridges, it is

found that there are a lot of data points that are geographically incorrect. A number of bridge records contain missing or invalid geographic coordinates, which causes some extreme outliers.

First, a substantial number of bridge records contain missing or invalid geographic coordinates. Some entries have NaN values for latitude and longitude, while others are assigned the default coordinates (0.0, 0.0). In terms of the What dimension, it is unclear what exact location is being captured, as essential spatial attributes are absent. This represents a failure of syntactic completeness and directly limits the usability of the data.

Secondly, multiple bridges appear far outside the geographical boundaries of Bangladesh. This indicates problems in the Where dimension. In addition to missing values, multiple bridges appear far outside the geographical boundaries of Bangladesh. Some are located in neighbouring countries or in water bodies. These extreme spatial outliers indicate a lack of semantic accuracy, as the data deviates significantly from reality. Such errors may stem from human data entry mistakes, including swapped latitude and longitude values or misplaced decimal points.

A major cause of these anomalies lies in the relational dependency between bridges and Location Reference Points (LRPs). If LRPs in the road dataset are incorrectly positioned, all bridges linked to those LRPs inherit the same spatial error. This illustrates how weaknesses in the How of data generation. Specifically, the process by which bridge locations are derived from LRPs propagate errors across datasets.

1.2.2 Topological and Relational Inconsistencies

Next to spatial and coordinate errors, the bridges dataset also contains topological and relational inconsistencies. Bridges are expected to be positioned on a road and linked to a valid LRP. However, a subset of bridges is assigned to LRPs that do not exist in the road dataset. These bridges are often flagged with an estimated location status of “error.” From the perspective of Marsden and Pingry (2018), this reflects a failure in the Which and How dimensions: the instruments or linking mechanisms used to connect bridges to roads do not function consistently across datasets.

From a pragmatic data quality perspective, this represents a critical issue. While a bridge may physically exist, the absence of a valid connection to a LRP in the road data makes it hard to position the road on the correct point. Such inconsistencies likely originate from differences in the When and How of data collection. If the road and bridge datasets were collected or updated at different times, discrepancies between LRPs may happen. According to Dong (2019), reproducibility requires explicit documentation of how relational links are constructed. In the absence of such documentation, systematic verification and cross-referencing with the road dataset are necessary to restore consistency and ensure the bridges dataset meets minimum data quality thresholds.

1.2.3 Attribute Outliers and Measurement Errors

The bridges dataset contains substantially more attribute information than the roads dataset, including bridge width, length, construction year, chainage, and estimated location type. So next to data quality errors in the geographical information, in this data there are more opportunities for data quality errors.

Several bridge attributes exhibit extreme outliers that are highly implausible. For example, a box culvert on LRP005b of road Z7456 is recorded with a width of 702 meters. This value is several orders of magnitude larger than typical box culvert widths, which generally range between 1 and 2.4 meters [1]. Such deviations indicate failures in syntactic accuracy, most likely caused by unit inconsistencies, misplaced decimal points, or manual entry errors.

From the Marsden and Pingry (2018) perspective, this represents a breakdown in the What and Which dimensions. The What captured in the dataset does not accurately reflect the physical characteristic of the bridge, and the measurement instrument or recording procedure, Which, may have been incorrectly applied. Unlike spatial errors, attribute outliers do not directly affect the geographic placement of bridges, but it can influence the analysis of the bridges. Therefore, these errors must be detected and corrected. As Dong (2019) argues, even when data are internally consistent, incorrect parameter values undermine the validity of the results. Therefore, detecting statistical outliers and verifying attribute plausibility are necessary steps to ensure that the dataset meets minimum standards for reliability and reproducibility.

2 Conceptual Cleaning Strategy

In this chapter, the conceptual strategy for solving each identified type of data quality issue is presented and discussed.

2.1 Road Dataset

To resolve the identified data quality issues in the roads dataset, an automated, algorithmic strategy is formulated. Given the size of the dataset (885 roads), manual correction is unfeasible.
To fix the extreme spatial outliers and coordinate errors found in the dataset, a two-step approach is used: first the errors are detected statistically, then they are geometrically corrected.

2.1.1 Detection: distance measurement and outlier thresholding

Since these errors show up as unusually long jumps between points, they are detected by measuring the distances between consecutive LRPs.

- Haversine formula: instead of using simple straight-line distance, the distance is calculated by using the Haversine formula. This is a mathematical equation that takes into account the spherical shape of the earth, the distance between two points, and the radius of the earth, thus calculating a more accurate distance [5].
- For each road, the median and standard deviation will be calculated for the segment lengths. The median instead of the mean is used, because the mean is be heavily affected by the large outliers, which may lead to some smaller outliers failing to be detected, as they are being 'shielded' by the larger ones.
- Outlier detection: a segement is marked as an outlier if it is longer than:

  $$\text{Threshold} = \text{Median} + (3 \times \text{STD})$$

  Using a factor of 3 (the Z-score) ensures that only segments that are far above normal distances are flagged as errors [4].

2.1.2 Correction: selective geometric snapping

Once an outlier is found, deleting it would break the sequence of LRPs. Instead, the point is moved to a likely correct location:

- Interpolation: The expected position of the point is calculated by taking the midpoint between its direct neighbours, except when these neighbours are also detected as outliers. In that case, the next point is taken as reference point for interpolation.
- Selective coordinate adjustment: Errors usually affect only Latitude or Longitude, rarely both. The differences between the current and expected coordinates is compared. The LRP is then snapped back along the smaller axis. This reduces unnecessary changes to the data.

2.1.3 Strategy for topological inconsistencies

Topological errors can be hidden by large outliers. For example, a 300 km jump inflates the standard deviation, so a smaller 5 km zigzag looks normal, especially on roads with a smaller number of LRPs. To solve this the the strategy is to work iteratively. Every iteration, all outliers are detected and snapped back to position. Then, the median and standard deviation is calculated again based on the updated distance dictionary. This is done until the algorithm converges and the list of outliers and their location stays exactly the same over multiple iterations. This ensures the final road network is free of both major jumps, and smaller topological errors.

2.1.4 Limitations

A number of outlier patterns are not correctly detected by the algorithm. In this section, we will discuss these. Firstly, when there are more than two consecutive outliers, as in 1 A and B, numerous problems arise. For A, the middle one is not detected as outlier, as the distance to its neighbours is relatively small. As both neighbours need to be far away to be classified as an outlier, all three outliers are not detected. This could be fixed by iterating over points until we find a distance that is similarly large, and denote that as the end of the outlier group. In B, a group of outliers is detected, but not snapped back correctly, since the interpolated

location is calculated based on the other outliers in the group. In C, the top LRP is the start or end of the road, and in this case it is impossible to know for certain if LRP1 or LRP2 would be the outlier, since there is no trend to interpolate from. D is a special case, which was present in the dataset. A part of the road was moved up, but since this included the beginning of the road, it was impossible to interpolate this to the right location.

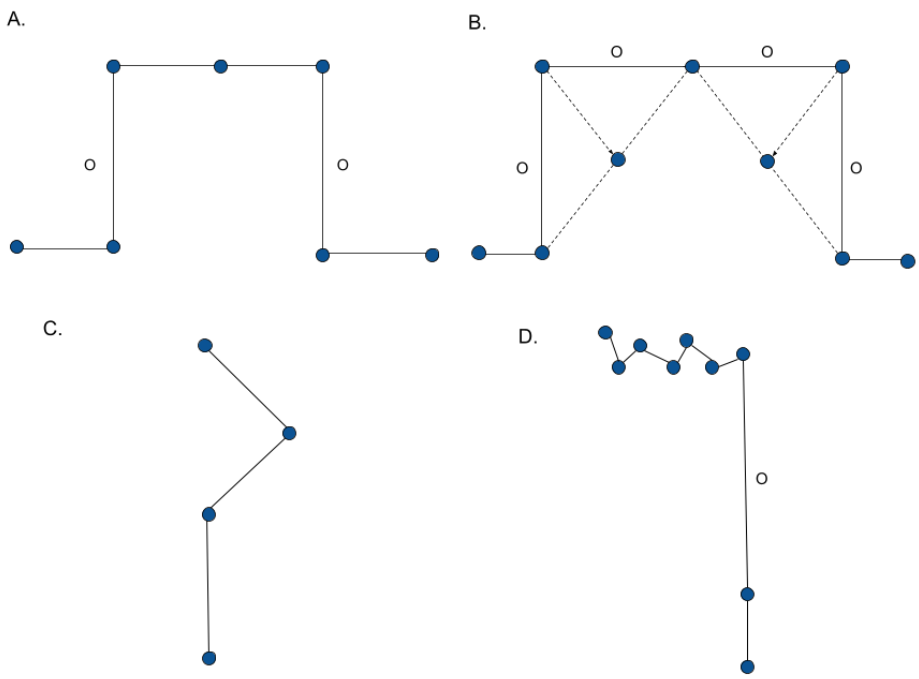

Figure 1: Various outlier patterns, 'o' means that edge is detected as outlier.

2.2 Bridges Dataset

This section outlines the conceptual strategy to obtain a cleaned and consistent bridges dataset. The strategy directly addresses the three previously identified problem categories: spatial outliers and coordinate errors, topological and relational inconsistencies, and attribute outliers and measurement errors.

2.2.1 Strategy for Spatial Outliers and Coordinate Errors

Filtering on borders and distance based validation
To identify bridges located outside Bangladesh, a polygon shapefile from Natural Earth Data will be used to represent the national boundary. Because Bangladesh has a complex shape and water boundaries may generate false positives, the cleaning will rely on distance-based validation rather than strict polygon containment. For each flagged bridge, the following diagnostic checks will be performed:

1. Check whether latitude and longitude are swapped.
2. Check for decimal errors (e.g., misplaced decimal points).
3. Check whether the associated LRP exists in the road dataset.
4. Check whether the bridge lies close to the road corresponding to its LRP.

If one of these issues explains the spatial deviation, the coordinate will be corrected accordingly.

Repositioning based on chainage
If no direct coordinate correction is possible, the bridge location will be reconstructed using the chainage, which is the distance between two points following the road.

This process is as follows:

1. Identify the road to which the bridge is assigned

2. Determine the two LRPs between which the bridge should lie, based on the chainage
3. Calculate the bridge location using (proportional) interpolation between LRP1 and LRP2
4. Base the coordinates on the relative chainage distance of the bridge along the road

This method ensures that even bridges with incorrect coordinates can be repositioned consistently along the road network.

2.2.2 Strategy for Topological and Relational Inconsistencies

The second part of the strategy addresses inconsistencies between the bridges dataset and the road dataset.

Missing or Non-Matching LRPs
If a bridge references an LRP that does not exist in the road dataset, the following steps will be taken:

1. Search for similarly named LRPs, on the same road
2. If the LRP is out of bounds for the road, check wether the LRP exists on a different but similarly named road
3. Verify wether the bridge chainage corresponds to any LRP chainage on the expected road

If no matching LRP can be identified, the bridge will be interpolated along the road using chainage values and nearest LRPs, as described in Section 3.1.2. And if neither chainage nor LRP provides a reliable reference, the bridge will be flagged for manual inspection.

Duplicated LRP Names and Bidirectional Bridges
Some bridges appear multiple times in the dataset because they represent traffic that goes both ways. These duplicates often have slightly different coordinates and can have different construction years, as it can be an update to the bridge.

The cleaning strategy will:

- Identify potential duplicates based on identical road ID, chainage, and similar names.
- Compare coordinate differences.
- Retain both records if they represent legitimate dual structures.
- Merge records if they are confirmed duplicates with no structural distinction.

Clear criteria will be defined to distinguish between true duplicates and legitimate parallel structures.

2.2.3 Strategy for Attribute Outliers and Measurement Errors

The third section focuses on non-spatial attribute cleaning.

Detection of Extreme and Invalid Values
For attributes such as width, length, and chainage, the following filters will be applied:

- Remove negative values
- Flag zero values where physical implausible
- Detect extreme outliers using statistical thresholds

For bridge width and length, outliers will be identified using either:

- Median ± three standard deviations, per bridge type
- Interquartile Range method

Using a threshold that is bridge type specific, it can be ensured that culverts, large river bridges, and other types are not incorrectly treated as outliers.

Bridge name standardization
Bridge names will be standardized to improve consistency:

- Remove trailing spaces and inconsistent capitalization.
- Harmonize slightly different spellings.
- Retain the most complete version of a name where true duplicates exist.

- Bridges without names may be left unnamed unless reliable external verification is feasible. Note that this is a lot of manual work.

2.2.4 Strategy for Tackling TU Delft adapted bridges based on non-clean data

As discussed in the document "EPA133A Assignment 1 – Data Quality", the file 'BMMS Overview.xlsx' was partially modified by TU Delft staff. This is visible in the column 'EstimatedLoc'. When this column contains the value 'road interpolate', it indicates that TU Delft staff adjusted the longitude and latitude of the bridge. This adjustment was based on the two closest LRPs to the bridge, using the chainage as a reference.

However, as previously discussed, the original LRP data is not clean. In the analysis, this data is cleaned before being used. Therefore, it is advisable to reapply the above-described method used by TU Delft staff, but this time using the cleaned LRP dataset.

The applied methodology was introduced in Section 2.2.1 and is described in more detail below.

- Filter the bridge dataset to include only bridges for which the longitude and latitude were adjusted by TU Delft based on LRP data.
- Link the chainage from the RMMS dataset to the corresponding LRPs in the cleaned roads dataset.
- Use an iterative procedure to determine between which two consecutive LRPs the bridge chainage (from BMMS Overview) is located, based on the cleaned LRP roads dataset.
- Calculate the relative position (fraction) of the bridge between the two LRPs (preceding and succeeding), based on their chainage values.
- Compute the new longitude and latitude of the bridge using linear interpolation:

$$\text{longitude} = \text{longitude}_{\text{before}} + \text{fraction} \cdot (\text{longitude}_{\text{after}} - \text{longitude}_{\text{before}})$$

$$\text{latitude} = \text{latitude}_{\text{before}} + \text{fraction} \cdot (\text{latitude}_{\text{after}} - \text{latitude}_{\text{before}})$$

- Update the calculated longitude and latitude in the BMMS Overview dataset.

3 Prioritization of Issues

In this chapter the prioritization of data quality issues is showed.

3.1 Road Dataset

3.1.1 Priority 1: Extreme Spatial Outliers

The highest priority is detecting and correcting extreme spatial outliers.They place road segments far from their true location, sometimes even in the ocean. These errors break the simulation, causing impossible travel times and preventing a connected road network. Removing these outliers first ensures the data is reliable for further cleaning.

3.1.2 Priority 2: Topological Inconsistencies

After the outliers, the zigzags and backtracking, where LRP_n is not followed by LRP_{n+1}.These errors distort distances and network flow, which can bias logistics calculations or create disconnected sections. Fixing them after outliers ensures accurate road geometry.

3.2 Bridges Dataset

3.2.1 Priority 1: Bridges outside Bangladesh

As discussed in Section 3.3.1, it is important to first address the most significant outliers. These concern bridges located outside the national borders of Bangladesh. They are therefore examined first and corrected where possible.

3.2.2 Priority 2: TU Delft adjusted data

As discussed in Section 2.2.4, TU Delft staff modified part of the bridge dataset based on a non-cleaned roads dataset. It is therefore important to reassess this bridge data and, where possible, recompute the coordinates using the cleaned roads dataset. The original, non-cleaned dataset may contain inaccuracies, which could affect the reliability of the adjusted bridge coordinates. By using the cleaned dataset, the longitude and latitude values of the bridges can be determined more accurately.

4 Implementation and Results

Implemented solutions (algorithms + results) are well presented and explained; clear link (reference) to the code submitted

4.1 Implementation and Results Roads dataset

The solution for cleaning the road datasets followed the conceptual strategy described above. For each road, a dictionary of distances between LRPs was built using the Haversine formula. Outliers were detected and corrected by interpolating their position based on their neighboring points, provided that those neighbors were not outliers themselves.

This process was run iteratively. In each iteration, a new dictionary of distances was calculated, along with updated statistical measures, allowing the detection of additional outliers that became apparent after earlier corrections. By repeating this, the datasets were gradually cleaned and brought into a consistent and reliable state.

The results show a significant improvement in data quality. For example, on Road N1, 16 outlier segments were initially detected, including extreme errors where points were hundreds of kilometers apart. After three iterations, all outliers were resolved, and the road data was fully consistent. Similarly, on Road N102, critical errors of over 200 kilometers were corrected, resulting in a fully streamlined dataset where all points now lie within reasonable distances from each other. The results can be seen in figure 2. It is clearly visible most big outliers are gone, as well as many smaller ones inside Bangladesh.

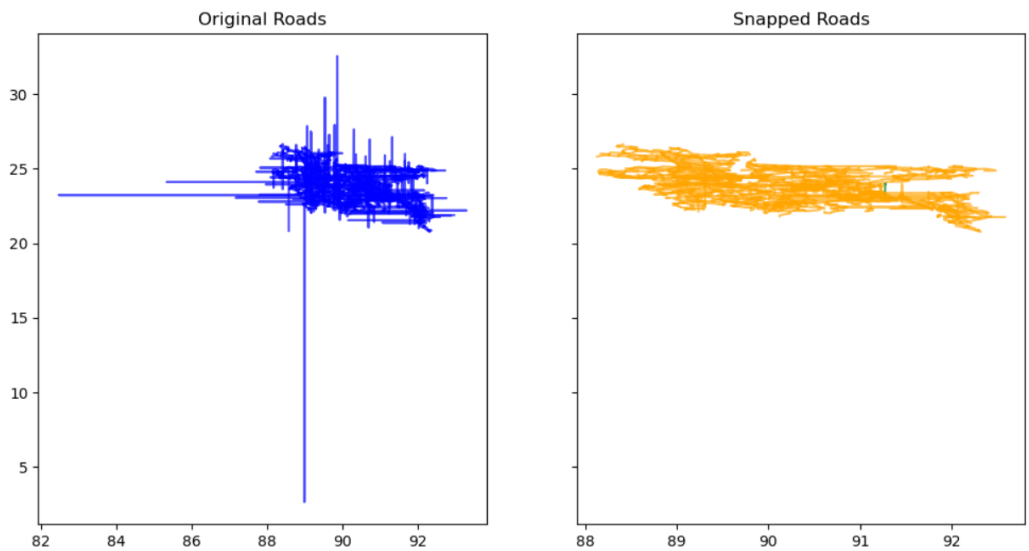

Figure 2: Dirty and cleaned roads.

4.2 Implementation Bridges dataset

4.2.1 Bridges outside Bangladesh

This section describes the implementation and results of the cleaning strategy for bridges located outside the national borders of Bangladesh.

To identify bridges located outside Bangladesh, a polygon shapefile from Natural Earth Data was used. Two boundary representations were initially considered: (1) national borders including maritime boundaries and (2) land borders only. Using borders that include water resulted in a large number of false positives, as bridges near the coast or river mouths were incorrectly classified as outside Bangladesh. Using land borders alone was also problematic, as these boundaries do not form a closed polygon and therefore cannot be used directly for point-in-polygon testing. To overcome this limitation, a distance-based validation approach was implemented instead of strict polygon containment.

First, the Bangladesh polygon was extracted from the Natural Earth Countries shapefile and projected with a coordinate reference system to enable distance calculations. All bridge coordinated were projected to the same CRS and for each bridge, the distance to the nearest Bangladesh border was computed. A bridge was

classified as being far outside Bangladesh if its distance to the boundary exceeded 5 kilometers. This threshold was chosen to avoid misclassifying bridges close to borders or coastlines, while still capturing clear spatial outliers. This procedure resulted in a subset of bridges that were clearly located well outside Bangladesh. These bridges formed the starting point for further diagnostic and corrective steps.

For bridges identified as far outside Bangladesh, the next step was to determine whether their spatial error could be explained and corrected using relational information. Each flagged bridge was matched to the road dataset using the combination of road ID and LRP name. If a matching LRP was found in the roads dataset, the bridge's latitude and longitude were replaced with the coordinates of the corresponding LRP. This step corrected a subset of bridges whose incorrect location was caused by erroneous bridge coordinates rather than incorrect road geometry. A Boolean was stored to indicate whether a successful LRP match was found.

After this first correction, a small number of bridges still appeared outside Bangladesh. Inspection of these remaining cases revealed a consistent pattern: the latitude values were unusually high, while the longitude values were unusually low, which is not plausible for Bangladesh. For a number of bridges, swapping latitude and longitude values placed the bridge correctly on a road segment within Bangladesh. This strongly suggests a human data entry error during data collection. For these cases, the latitude and longitude were swapped programmatically, after which the bridges were no longer classified as outside the national boundary.

After applying both corrections and coordinate swapping, a limited number of bridges remained outside Bangladesh. Most of these bridges did not have a matching LRP in the road dataset, making automated correction unreliable. In accordance with the conceptual strategy, these bridges were left unmodified and implicitly flagged for manual inspection.

4.2.2 TU Delft adjusted data

For the implementation of the TU Delft–adjusted data, the methodology described in Section 2.2.4 is applied. First, the chainage values of the LRPs are added to the roads dataframe by merging the roads and LRP dataframes. Subsequently, all bridge observations for which the location was adjusted by TU Delft are filtered from the bridges dataframe, as indicated in the "EstimatedLoc" column.

Next, the preceding and succeeding LRPs in the roads dataset corresponding to each bridge LRP are identified, based on chainage and the associated road segment. If no valid preceding or succeeding LRP is found, the bridge is excluded from further recalculation.

Finally, the relative position (fraction) of the bridge between the two LRPs is computed. This fraction is then used in the interpolation formula below to determine the updated latitude and longitude of the bridge.

$$\text{longitude} = \text{longitude}_{\text{before}} + \text{fraction} \cdot (\text{longitude}_{\text{after}} - \text{longitude}_{\text{before}})$$

$$\text{latitude} = \text{latitude}_{\text{before}} + \text{fraction} \cdot (\text{latitude}_{\text{after}} - \text{latitude}_{\text{before}})$$

4.2.3 Unaddressed Aspects

Due to time constraints, several components outlined in section 2 "Conceptual Cleaning Strategy" were not implemented. These include: the identification of missing or non-matching LRPs, the detection of duplicated LRP names and bidirectional bridges, the detection of extreme and invalid values, and bridge name standardisation.

Additionally, not all bridges could be relocated based on chainage. This was primarily due to inconsistencies between the LRP–road combinations assigned in the Bridges dataset and those in the cleaned roads dataset. While chainage-based interpolation (Section 2.2.4) resolved most cases, relocation remained impossible for bridges whose assigned LRPs lacked a valid preceding or succeeding LRP on the specified road segment.

4.2.4 Results

Extreme spatial outliers were robustly identified using distance-based validation rather than strict polygon containment. A significant portion of mislocated bridges was successfully corrected by linking them to cleaned road LRPs. Additional errors caused by swapped latitude and longitude values were detected and corrected. Bridges that could not be reliably corrected due to missing relational information were retained but identified as unresolved. This approach balances automated correction with data integrity by avoiding speculative adjustments when insufficient reference information is available.

It is difficult to conclusively determine from the results whether the newly assigned locations are fully accurate. However, clear differences are observed before and after the implementation of the algorithm, and the majority of previously identified outliers appear to have been corrected.

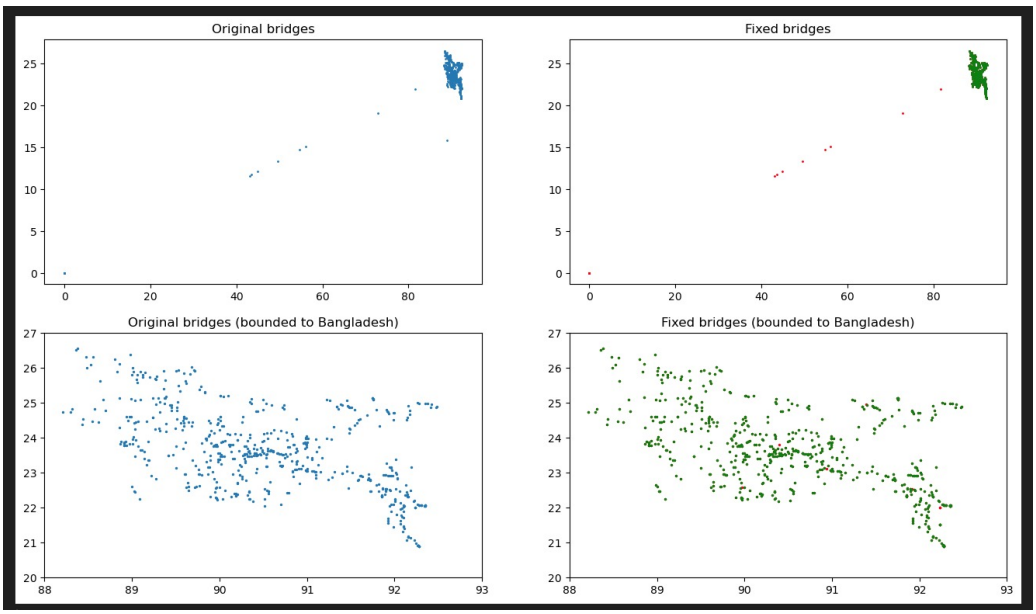

Figure 3: Dirty and cleaned bridges.

5 Discussion and Reflection

The implemented solution has a number of limitations. For the roads, several of them were mentioned in the conceptual cleaning strategy. The implementation of this strategy brought some new limitations. for example, if the neighbour of an outlier LRP was also an outlier, the algorithm was supposed to iterate untill it found the next LRP that was not an outlier. However, in the implementation only one cycle of this iteration was performed, meaning htat a neighbour would be at most two away from the first LRP. Extending this iteration would be a good improvement. Furthermore, a solution that would find a distance outlier and then iterate over subsequenct distances until it finds a distance that is almost the same, and marks all of the in-between points as outliers would be better at dealing with situation A & B in 1. Lastly, some manual inspection and correction would be neccesary for both the roads and bridges, as it is impossible to automatically infer their location from other data.

A central choice in the bridge cleaning process was the use of a 5 km distance threshold to identify bridges located outside Bangladesh. While this approach proved effective for detecting extreme spatial outliers, the threshold remains inherently subjective. Different values would lead to different classifications, illustrating that data cleaning introduces assumptions into the How of data processing, as described by Marsden and Pingry (2018). These assumptions should be explicitly acknowledged when interpreting simulation results.

The cleaning of the bridges dataset was strongly dependent on the quality of the road dataset, particularly the Location Reference Points (LRPs). Bridge locations were corrected using LRP matching and interpolation along cleaned road geometries. As a result, any remaining inaccuracies in the road dataset may propagate into the bridges dataset. This highlights a hierarchical dependency between datasets and underlines the importance of addressing road-level errors prior to bridge-level corrections.

Another important consideration is the risk of overcorrection. Automated procedures were used to replace bridge coordinates based on relational information and to correct suspected latitude–longitude swaps. While these corrections were supported by clear spatial patterns, automated cleaning cannot fully distinguish between true errors and rare but valid edge cases, particularly near national borders. To mitigate this risk, bridges that could not be reliably corrected were left unchanged and flagged, prioritizing data integrity and transparency over completeness, in line with Dong (2019).

Finally, the distance-based border validation approach is well suited for identifying extreme spatial anomalies but does not capture more subtle positional errors, such as bridges located within Bangladesh but incorrectly aligned with their associated roads. Consequently, the implemented strategy focuses on improving structural validity rather than achieving perfect spatial precision.

Overall, this demonstrates that data cleaning is an assumption-driven process whose choices directly influence model validity. Transparent documentation of these choices is therefore essential for reproducibility and meaningful simulation outcomes.

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Acknowledgements

5.1 AI Statement

This report was developed with the support of AI tools. However, all main ideas, prompts, and inputs came from the authors. AI was used only as a supporting tool during the development of this report, mainly in two ways:

- Coding: AI was used to help solve coding problems when the team encountered errors they could not fix on their own. It was also used to brainstorm strategies for improving data quality. This helped the team to work more efficiently.
- Writing the report: AI was also used during the writing phase of the report. It helped improve the flow and clarity of the text and supported the authors in making the document more structured and consistent. AI suggested improvements to sentence structure, wording, and transitions between sections, which helped create a clearer report. Despite this support, the content, analysis, and conclusions of the report remain the work of the authors.document.

5.2 Group Statement

Underneath, the role and contribution of each group member is described. Overall, the assignment was completed as a group effort, and everyone contributed to the final version of the report.

- Niels van den Dool: Niels was mainly responsible for the coding. He focused on fixing the road dataset and worked on improving it during the project. He also finalized both the road and bridges datasets for the final deliverable.
- Eli Kapteijn: Eli worked together with Niels on the road dataset algorithm. In addition, he wrote parts of the road dataset in the report and contributed to explaining this part of the project.
- Nick Schreurs: Nick worked on the bridges dataset for and reviewed its contents. He also wrote several parts about the bridges dataset in the main report.
- Lieke Wahlen: Lieke worked on the bridges dataset together with Nick. She wrote multiple sections about the bridges dataset in the report and contributed to describing this part of the work.
- Mirte Wildeboer: Mirte carried out a first exploration of the bridges file and algorithm. She also created the structure of the report and wrote the sections about the road dataset.

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